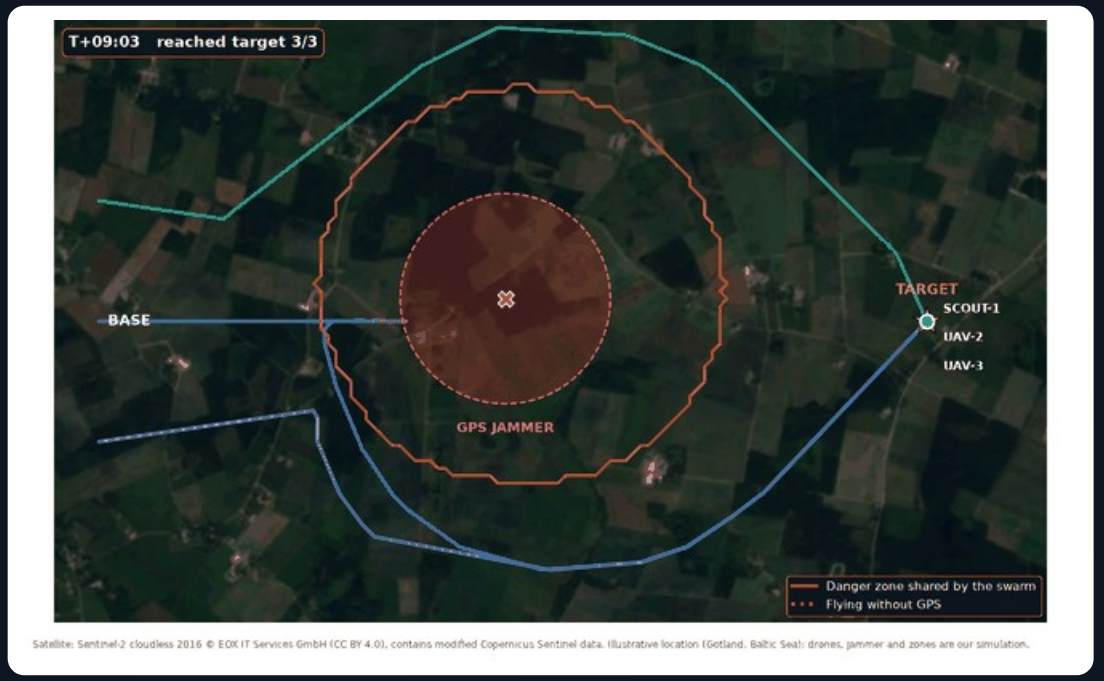
EUDIS DEFENCE HACKATHON 2026 · CHALLENGE 1

One drone gets jammed — the whole swarm learns

GPS-denied navigation that detects the attack, shares it, and still completes the mission



Our swarm mission on a real satellite map (animated). Illustrative location: Gotland, Baltic Sea.

Satellite: Sentinel-2 cloudless 2016 © EOX IT Services GmbH (CC BY 4.0), contains modified Copernicus Sentinel data

Den cyber security **Guled** computer engineering

GPS is attacked “really often” — drones land blind

RELEVANCE

What the Swedish Air Force (Helicopter Wing) told us

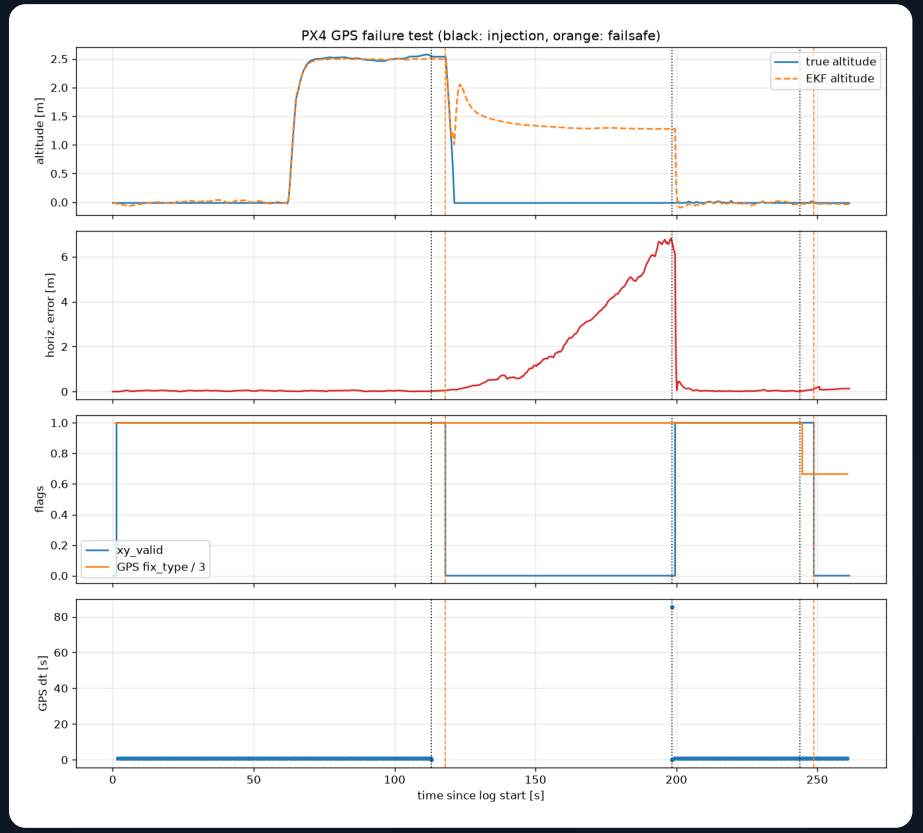
“It happens really often”
Jamming and spoofing — widely reported over the Baltic

“Detect it as soon as possible”
Both jamming and spoofing, depending on the mission

“Snow hides the landmarks”
Pre-made maps stop matching in winter

“A drone once built its own map to get home”
That real case is our starting point

Public evidence: GNSS interference over the Baltic is mapped every day from airliner data (gpsjam.org).



5 s
autopilot reaction time

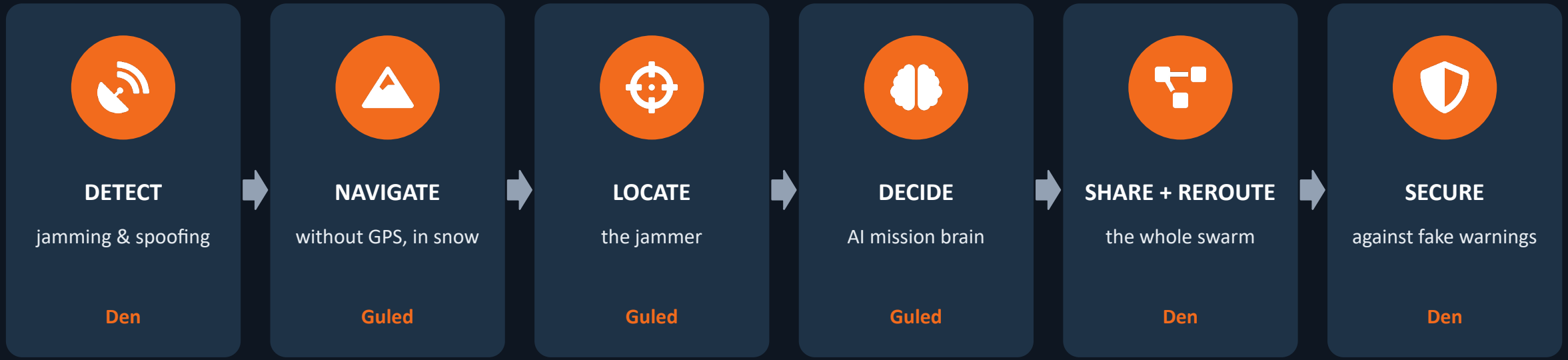
blind
landing — never flies home

The military told us what is really missing

RELEVANCE

“Detect and return home — we already have that. ***We need the drone to tell the others, and the swarm to find its own way to the target.”***

Military end-users, Friday — this changed our project



Detect: jamming in 1 s, slow spoofing in 13 s

INNOVATION

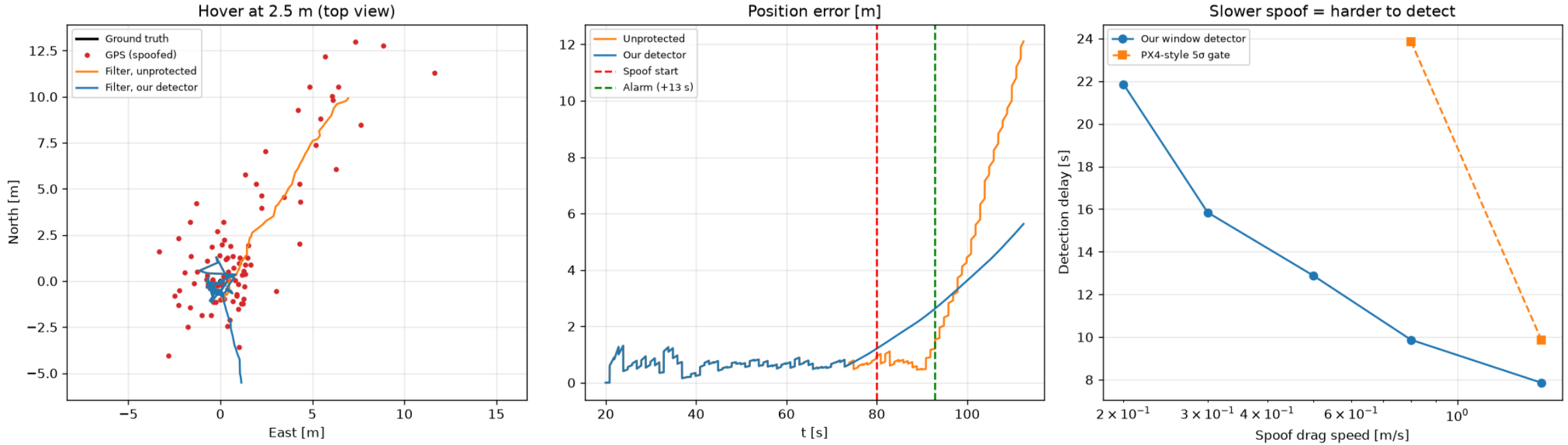
1 s vs 5 s
jamming: our monitor vs PX4 (live)

13 s
0.5 m/s spoof caught (real flight data)

never
PX4-style check notices that spoof

2 / 50
false alarms without attack

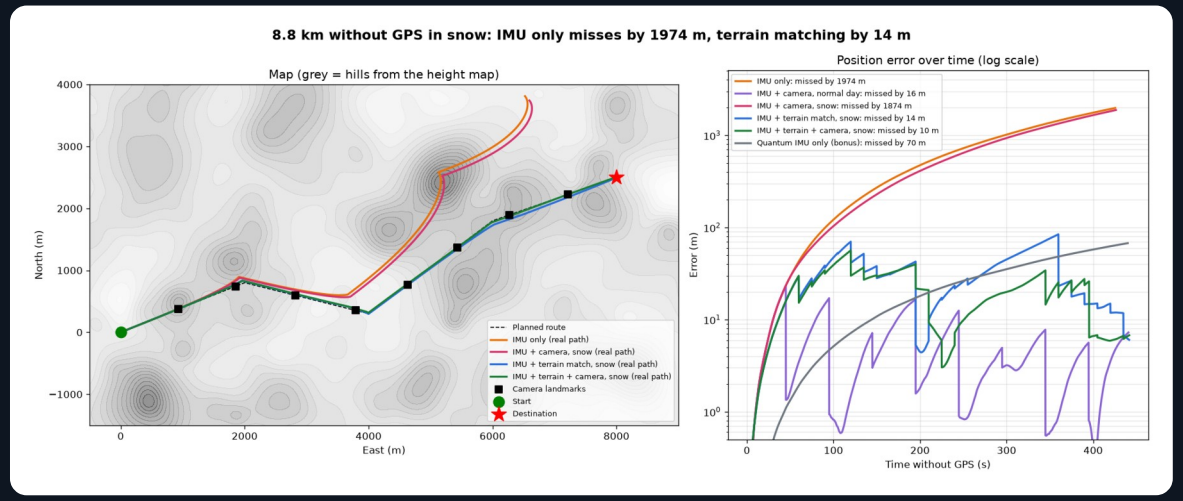
GPS noise is random — a spoofer always pushes the same way. We compare GPS with the drone's own motion sensors (IMU) over a 20 s window and catch the push.



Navigate in snow — then find the jammer

INNOVATION

NAVIGATE · 8.8 km without GPS, snowy day



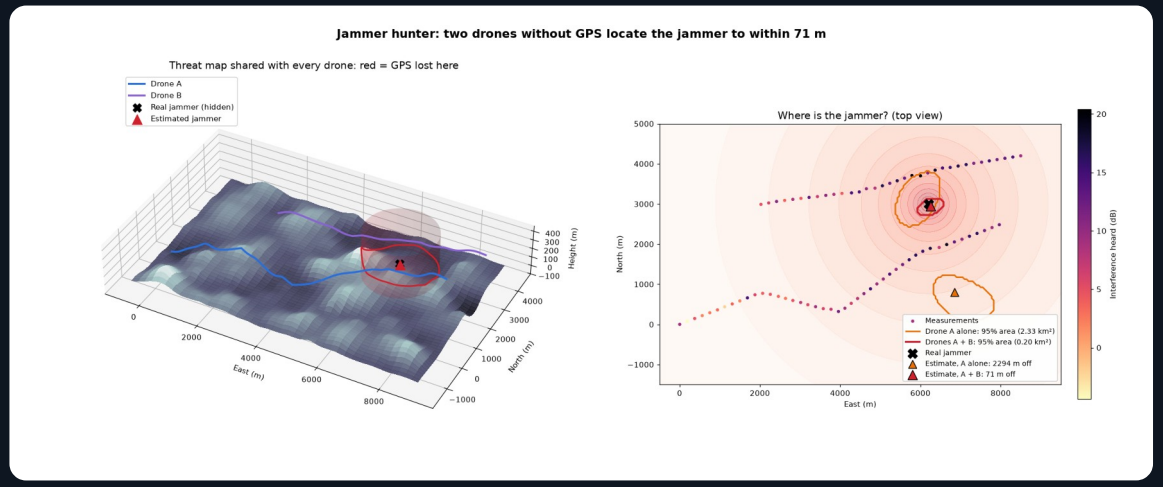
1 974 m
off: IMU only

1 874 m
off: camera in snow (0/8 landmarks)

14.5 m
off: terrain matching

Laser altimeter vs height map: the shape of the hills survives snow. Re-tested on the REAL Bråviken height map (SRTM) — see slide 10.

LOCATE · every jammer gives itself away



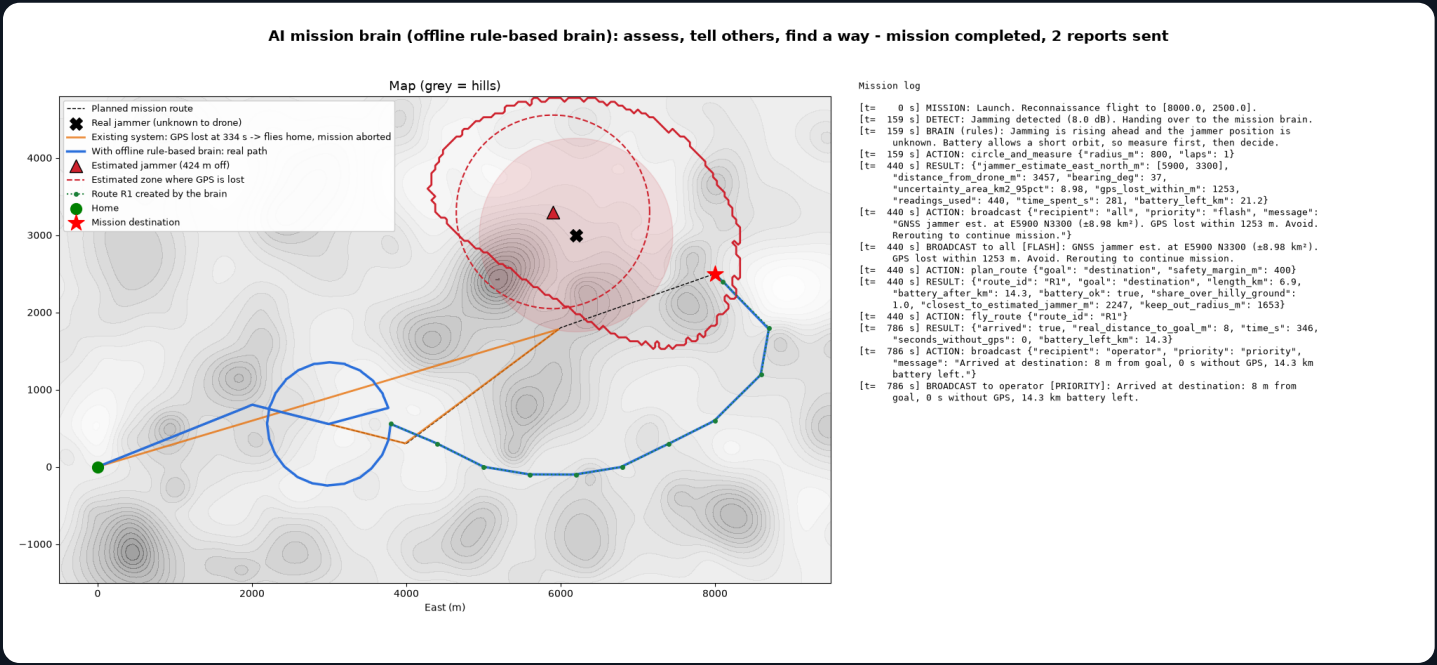
~300 m
one drone (typical)

~120 m
two drones (typical, 10 runs)

GPS receivers measure interference — louder means closer.

Decide: an AI mission brain that can't break the rules

INNOVATION



Shown: offline rule-based mode. Claude mode uses the same tools and safety checks.

AI decides what
Assess, measure, plan — explains each decision in plain language

Tested code decides how
4 safe tools: measure, plan route, fly, broadcast

Safety check on every action
Battery reserve and limits — the AI never flies the drone

aborted
today: GPS lost → flies home

8 m
from target: mission completed

One scout gets jammed — the swarm gets through

INNOVATION



Flight-radar replay of our swarm simulation — drone expert Isabella's idea, built by Guled

0 / 3
reach the target today

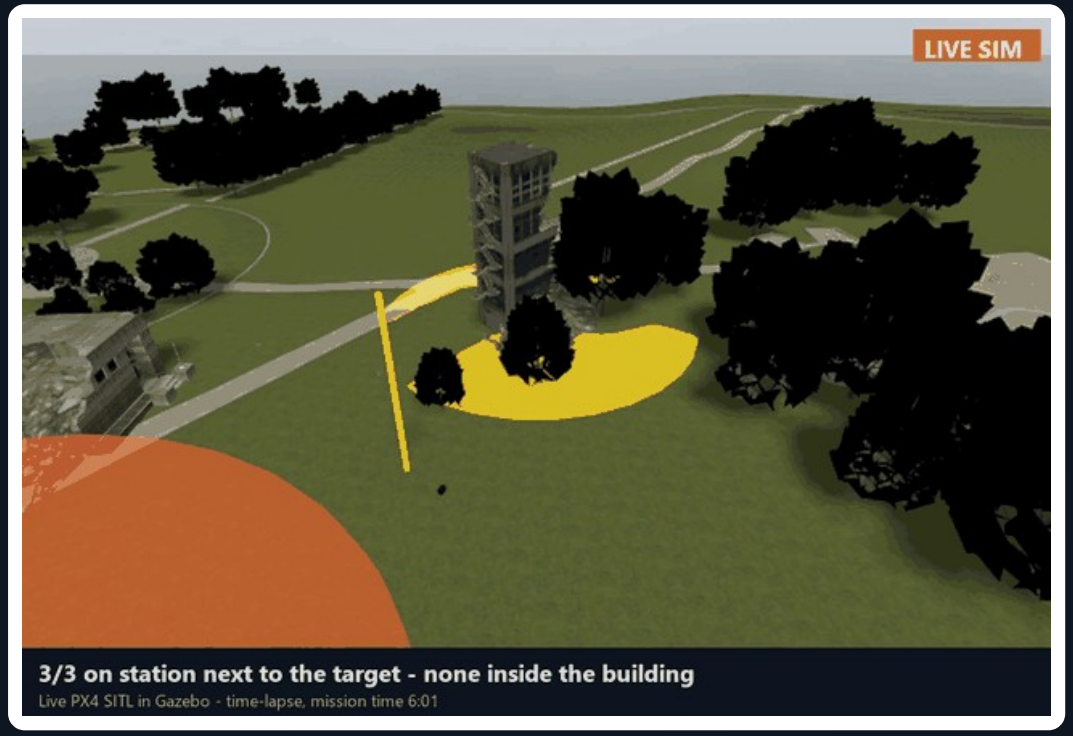
24 / 24
with our swarm (8 random runs)

16 s vs 68 s
without GPS: swarm vs each alone

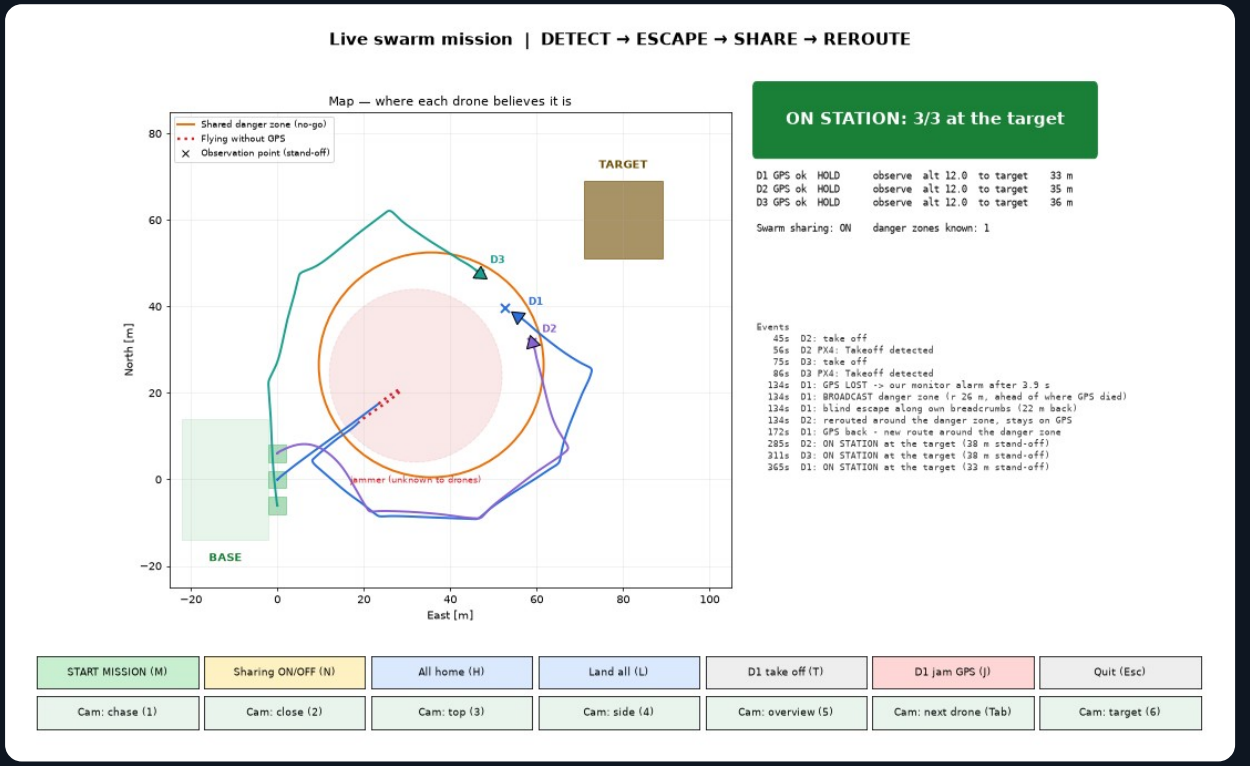
−16 %
mission time

Live: 3 real PX4 autopilots fly the mission in Gazebo

PROGRESS



Recorded live from Gazebo (time-lapse). Sharper video: media/gazebo_mission.mp4



Our mission app: every drone's position, GPS state and the shared danger zone

1 / 3
drones ever jammed (the scout)

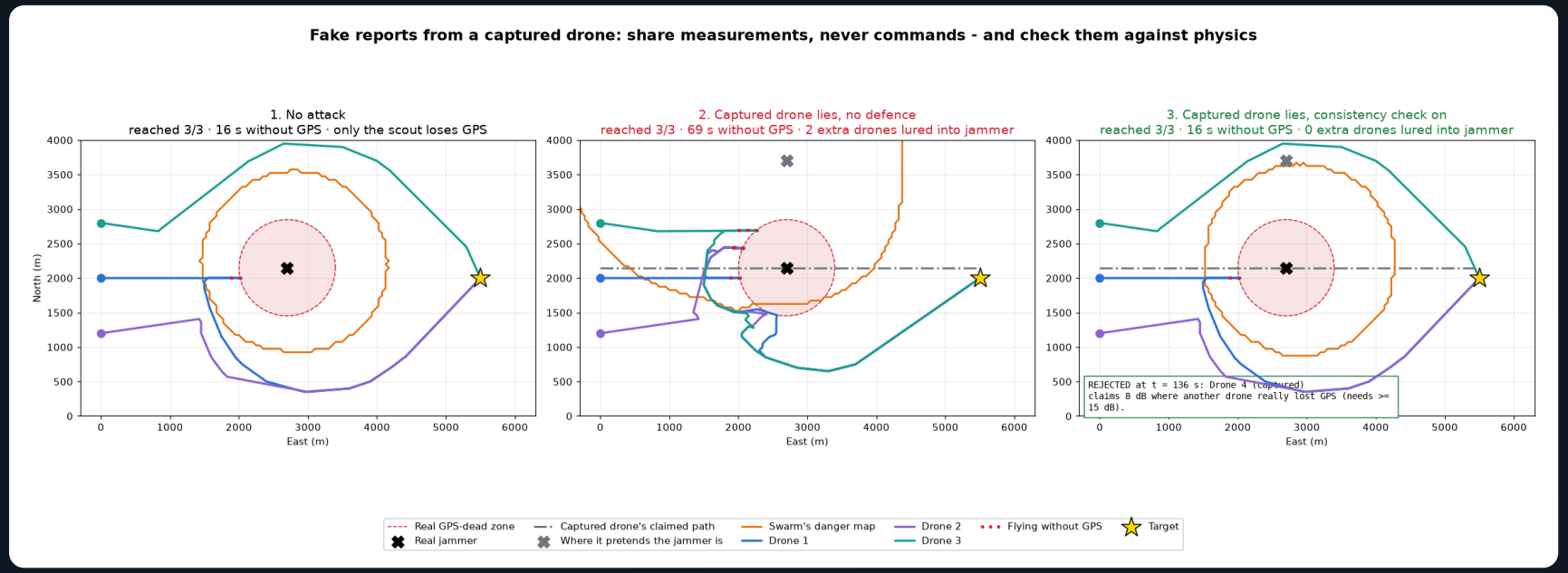
3 / 3
on station at the target

33–38 m
stand-off, never inside the building

7
live test runs to get it working

We attacked our own system: fake warnings

INNOVATION



9 → 1
drones lured into the jammer: no defence vs ours (6 runs)

6 / 6
times the liar was caught

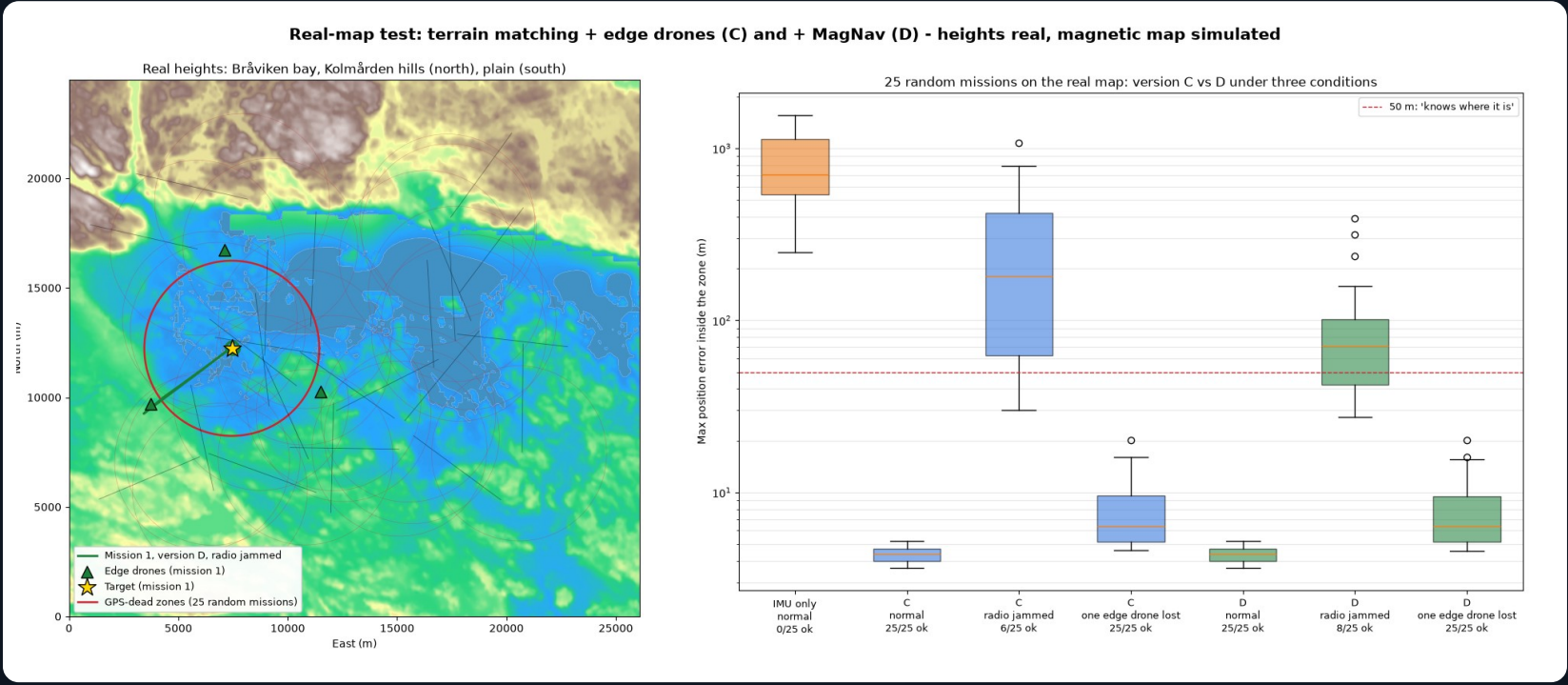
0
honest drones wrongly rejected

"A warning is cheap to believe. An all-clear must be proven."

Share measurements, never commands · check every report against physics and real GPS-loss evidence

Real terrain: 25 missions over Bråviken, Norrköping

PROGRESS



Built by Guled · navigation/real_map/ · heights real (NASA SRTM via AWS Terrain Tiles), magnetic map simulated

Honest finding: real ground is flatter than our simulated hills, and ~21 % of each route is over water. With the drones' radio link jammed too, terrain matching often has nothing to recognise — MagNav halves the error, but it is not a full fix yet.

25 / 25
within 50 m, radio link working (~4 m typical)

0 / 25
IMU only (~700 m typical)

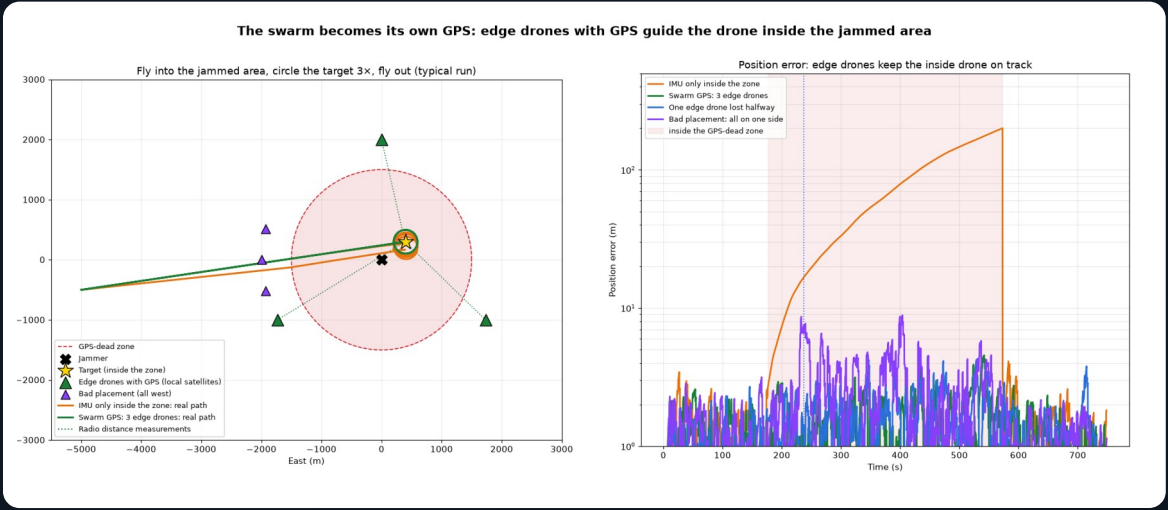
180 → 71 m
radio jammed too: terrain only vs + MagNav (typical)

6 → 8 / 25
radio jammed too: success

Target inside the zone? The swarm is its own GPS

INNOVATION

SWARM GPS · edge drones keep GPS and range the drone inside

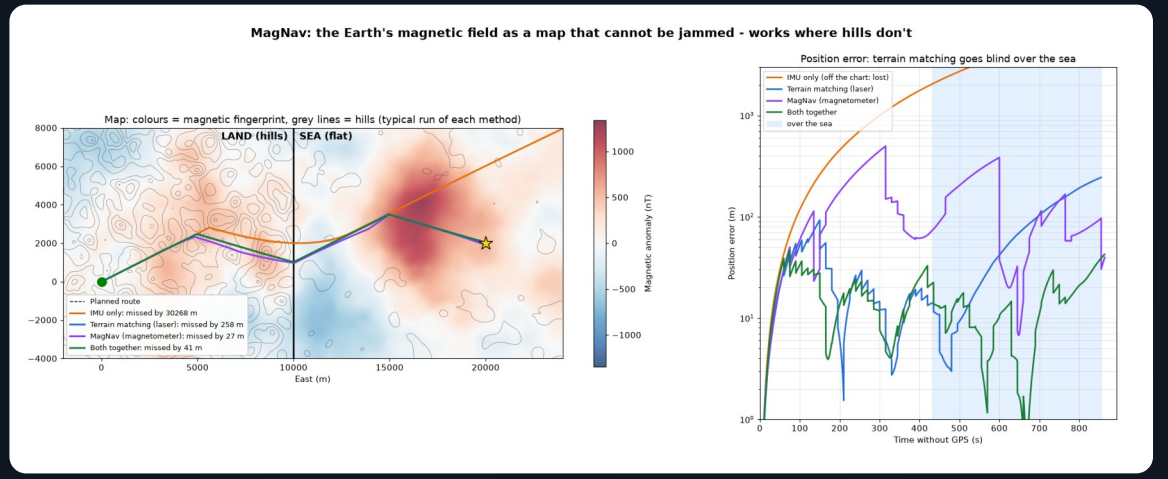


59 m → 1 m
error while observing the target (20 runs)

2 m
still, if one edge drone is lost

Radio ranging on the drones' own link — not the band a GPS jammer hits.

MAGNAV · the Earth's magnetic map can't be jammed



202 m
terrain matching — blind over the sea

38 m
terrain + MagNav (typical, 20 km)

It only listens — nothing to jam or spoof. Works over the Baltic, at night, in snow.

48 hours of progress — and what's next

PROGRESS

4
expert inputs

17
logged iterations

31
commits on GitHub

WHY IT GETS ADOPTED

Software only: runs on drones that already fly PX4 — no new hardware to buy

Users: armed forces and drone makers building on PX4; procurement via FMV

Dual use: civil drones and aviation in the jammed Baltic

Path: simulation → Terra lab → field test → PX4 plug-in

Honest limits
Simulation so far (navigation re-tested on real Bråviken terrain) · jammer model is free-space · the swarm assumes its radio link survives · very slow spoofs (< 0.1 m/s) are not caught

NEXT



Real drones in Terra lab
Holybro PX4 drones + Vicon ground truth at LiU



Real jamming indicators
u-blox receivers already report interference to PX4



AquaGuard Technologies
Satellite/SAR imagery as backup map; our jammer reports as their data feed



Operator in the loop
Secure drone-to-drone link, human approval for reroutes

Team

TEAM



Den

Cyber security student

Detection · swarm sharing · attack defence · live PX4 mission



Guled

Computer engineering student

GPS-free navigation · jammer hunter · AI mission brain · slides

Shaped by: Swedish Air Force Helicopter Wing · military end-users · drone expert Isabella · a satellite company · fellow participants

One drone gets jammed — the whole swarm learns.
If the enemy fakes the warnings, physics exposes the liar.